

SO101 3-DOF Arm: Simulation Model and Teaching Kinematics

A self-contained teaching note

1 Purpose

This document explains the current 3-degree-of-freedom arm project from first principles. It is written for a reader who has not seen the code before.

The project intentionally has two kinematic layers:

1. An intact MuJoCo simulation model. This preserves the recognizable SO101 visual structure and uses the STL mesh assets.
2. A clean Denavit-Hartenberg teaching model. This gives a compact parameter table, forward kinematics, Jacobian, and inverse kinematics.

The MuJoCo model is the visual and simulation model. The DH model is the mathematical teaching model. They serve different purposes and should not be forced into a single model by moving mesh geometry into DH frames.

2 Abbreviations

Abbreviation	Meaning
3-DOF	Three degrees of freedom, meaning three independent joint variables
SO101	The original arm family from which the geometry/assets were derived
MJCF	MuJoCo XML format, the model-description format used by MuJoCo
MuJoCo	Multi-Joint dynamics with Contact, the physics simulator
STL	Stereolithography mesh file, used here for visual robot parts
TCP	Tool center point; here this is the <code>gripperframe</code> site used in FK/IK
EE	End effector, the final fixed tool and gripper subassembly
DH	Denavit-Hartenberg, a standard robot-link parameter convention
FK	Forward kinematics, computing pose from joint angles
IK	Inverse kinematics, computing joint angles from a desired pose/position
COM	Center of mass

3 Coordinate convention

The simulator uses a right-handed world coordinate system. In the project:

- x points forward,
- y points left,

- z points upward.

The joint vector is

$$\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} \text{shoulder_pan} \\ \text{shoulder_lift} \\ \text{elbow_flex} \end{bmatrix}.$$

4 The intact MuJoCo model

The MuJoCo body tree is:

world $\rightarrow$ base $\rightarrow$ shoulder $\rightarrow$ upper_arm
 $\rightarrow$ lower_arm $\rightarrow$ wrist $\rightarrow$ gripper $\rightarrow$ moving_jaw_so101_v1.

Only the first three joints are actuated:

Index	Joint name	Type	Range
1	shoulder pan	revolute hinge	$[-1.91986, 1.91986]$ rad, $\pm 110.0^\circ$
2	shoulder lift	revolute hinge	$[-1.7453293, 1.7453293]$ rad, $\pm 100.0^\circ$
3	elbow flex	revolute hinge	$[-1.69, 1.69]$ rad, $\pm 96.8^\circ$

The wrist and gripper bodies remain in the XML as fixed structure. The joint count is still three, but the visual and collision geometry keeps the original SO101 look.

The relevant fixed offsets in the MuJoCo body tree are:

From	To	Translation in parent frame [m]
base	shoulder	(0.0388353, 0, 0.0624)
shoulder	upper_arm	(-0.0303992, -0.0182778, -0.0542)
upper_arm	lower_arm	(-0.11257, -0.028, 0)
lower_arm	wrist	(-0.1349, 0.0052, 0)
wrist	gripper	(5.55112e-17, -0.0611, 0.0181)
gripper	gripperframe site	(0.012, -0.000218, -0.098127)

These are exact simulator offsets, not the simplified DH parameters. They keep the imported STL parts visually coherent.

5 Rigid transforms

A rigid transform is represented by a homogeneous matrix

$$\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ 0 & 1 \end{bmatrix},$$

where $\mathbf{R}$ is a 3×3 rotation matrix and $\mathbf{p}$ is a position vector.

For a revolute joint about the local z -axis,

$$\text{Rot}_z(q) = \begin{bmatrix} \cos q & -\sin q & 0 & 0 \\ \sin q & \cos q & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The exact MuJoCo FK is therefore a product of fixed body transforms and three joint rotations:

$$\mathbf{T}_{0,\text{ee}}(\mathbf{q}) = \mathbf{T}_{0,b} \mathbf{T}_{b,s}^0 \text{Rot}_z(q_1) \mathbf{T}_{s,u}^0 \text{Rot}_z(q_2) \mathbf{T}_{u,l}^0 \text{Rot}_z(q_3) \mathbf{T}_{l,w}^0 \mathbf{T}_{w,g}^0 \mathbf{T}_{g,\text{ee}}^0.$$

In the code, FK and IK use the `gripperframe` site as the end-effector reference point.

6 Why a separate DH model exists

The STL meshes and SO101 body frames were not authored as textbook DH frames. If the XML body tree is forcibly rewritten as DH frames, the visual mesh origins no longer line up and the robot can look disjointed.

For teaching, a separate DH model is cleaner:

- the table is short,
- FK is a loop over rows,
- the Jacobian is easy to derive,
- IK can be taught using planar two-link geometry after the base yaw.

7 Standard DH convention

This project uses standard DH:

$$\mathbf{T}_{i-1,i} = \text{Rot}_z(\theta_i) \text{Trans}_z(d_i) \text{Trans}_x(a_i) \text{Rot}_x(\alpha_i).$$

The four DH symbols mean:

Symbol	Meaning
θ_i	joint rotation about z_{i-1} ; here this is q_i
d_i	offset along z_{i-1}
a_i	link length along x_i
α_i	link twist about x_i

8 Deriving the DH parameters

The DH table is a teaching simplification of the intact robot. The goal is to keep the original 3-link arm structure while making the math transparent.

The derivation is easiest to read from the joint axes:

1. Joint 1 is the base yaw. Its axis is vertical, so we choose z_0 along the shoulder pan axis. The next joint axis is orthogonal to it, which gives the standard DH twist $\alpha_1 = +\pi/2$.
2. The shoulder pitch and elbow pitch are parallel hinge axes. The distance between their axes along the arm centerline is 117.5 mm, so the second link length is $a_2 = 0.1175$ m and $\alpha_2 = 0$.
3. The elbow pitch to fixed tool/wrist line is another parallel pair with a centerline separation of 95.0 mm, so $a_3 = 0.0950$ m and $\alpha_3 = 0$.
4. The shoulder axis sits 59.5 mm above the world origin, which is captured by $d_1 = 0.0595$ m. The remaining links are modeled as planar offsets, so $d_2 = d_3 = 0$.
5. The gripper/tool center lies another 10 mm along the final x -axis, which becomes the teaching tool offset $a_{tcp} = 0.0100$ m.

In other words, the DH model keeps the arm's three active axes and the gross link lengths, but it absorbs the detailed wrist and gripper shape into a fixed tool offset.

9 Teaching DH table

The teaching DH model uses:

i	Joint	θ_i	d_i [m]	a_i [m]	α_i
1	shoulder pan	q_1	?	?	?
2	shoulder lift	q_2	?	?	?
3	elbow flex	q_3	?	?	?

The TCP/tool offset is

$$a_{tcp} = ? \text{ m}$$

along the final x -axis. This represents the 10 mm ball radius/tool offset in the teaching model.

10 DH forward kinematics

The full DH transform is

$$\mathbf{T}_{0,tcp}^{DH}(\mathbf{q}) = \mathbf{T}_{0,1}(q_1)\mathbf{T}_{1,2}(q_2)\mathbf{T}_{2,3}(q_3) \text{Trans}_x(a_{tcp}).$$

For the TCP position, this can be reduced to a set of trigonometric equations for x , y , and z in terms of the joint angles and link lengths. (Derive these as part of your exercise!)

11 Workspace limits

For the DH teaching model, the moving part after the base yaw is a planar two-link arm:

$$L_2 = a_2 = 0.1175 \text{ m}, \quad L_3 = a_3 + a_{tcp} = 0.1050 \text{ m}.$$

Ignoring joint limits, the TCP distance from the shoulder pitch frame must obey

$$|L_2 - L_3| \leq \sqrt{x^2 + y^2 + (z - d_1)^2} \leq L_2 + L_3.$$

Numerically:

$$0.0125 \text{ m} \leq \sqrt{x^2 + y^2 + (z - 0.0595)^2} \leq 0.2225 \text{ m}.$$

So the idealized maximum reach from the shoulder pitch frame is

$$0.2225 \text{ m} = 222.5 \text{ mm}.$$

The base yaw sweeps this planar workspace around the vertical axis, but only through its joint range:

$$q_1 \in [-1.91986, 1.91986] \text{ rad} = [-110.0^\circ, 110.0^\circ].$$

Without joint limits, the TCP height is bounded by

$$z_{\min} = 0.0595 - 0.2225 = -0.1630 \text{ m}$$

and

$$z_{\max} = 0.0595 + 0.2225 = 0.2820 \text{ m}.$$

With the actual joint limits,

$$\begin{aligned} q_2 &\in [-1.7453293, 1.7453293] \text{ rad} = [-100.0^\circ, 100.0^\circ], \\ q_3 &\in [-1.69, 1.69] \text{ rad} = [-96.8^\circ, 96.8^\circ], \end{aligned}$$

the real reachable region is a subset of that ideal volume. A target can be inside the ideal distance bound and still fail if it requires a forbidden joint angle.

For inverse kinematics inputs, it is often useful to know approximate axis-aligned bounds of reachable TCP positions. With the DH teaching model and the joint limits above, a grid/optimization search gives:

$$\begin{aligned} x &\in [-0.1254, 0.2225] \text{ m}, \\ y &\in [-0.2225, 0.2225] \text{ m}, \\ z &\in [-0.1630, 0.2820] \text{ m}. \end{aligned}$$

Equivalently, in millimeters:

$$\begin{aligned} x &\in [-125.4, 222.5] \text{ mm}, \\ y &\in [-222.5, 222.5] \text{ mm}, \\ z &\in [-163.0, 282.0] \text{ mm}. \end{aligned}$$

These are independent coordinate bounds, not a rectangular box in which every point is reachable. A target such as (0.22, 0.22, 0.28) m is inside the coordinate ranges but is still unreachable because it violates the arm's radial reach. The negative x bound is possible because the signed planar term r can become negative when the elbow folds back within its joint limits.

12 Jacobian

The position Jacobian maps joint velocity to TCP linear velocity:

$$\dot{\mathbf{p}}_{tcp} = \mathbf{J}_p(\mathbf{q})\dot{\mathbf{q}}.$$

For the DH teaching model, derive the analytical Jacobian by taking the partial derivatives of your forward kinematics position equations with respect to each joint variable.

13 Inverse kinematics

The IK problem asks for $\mathbf{q}$ given a target end-effector position

$$\mathbf{p}_d = [x_d, y_d, z_d]^T.$$

Hint: The base joint q_1 can be found from the horizontal projection of the target.

Then the remaining two joints solve a planar two-link problem. You can use the law of cosines to find the two possible elbow branches for q_3 , and then solve for q_2 . (Derive the exact equations as part of your exercise!)

14 Validation

Current checks:

- The intact MuJoCo model loads and steps with three joints and three actuators.
- Exact simulator FK is checked by reading the `gripperframe` site after setting joint values.
- The IK solver uses an analytical derivation based purely on the DH parameters.

The important interpretation is:

- MuJoCo FK is exact for the simulation model.
- DH FK/IK is exact for the teaching model.
- They are intentionally separate so the visual robot stays intact.

15 Keyframes

The model includes:

Keyframe	Joint values [rad]	Joint values [deg]
home	[0, 0, 0]	[0.0, 0.0, 0.0]
ready	[0, 0.6, -0.8]	[0.0, 34.4, -45.8]
reach forward	[0, 1.0, -1.0]	[0.0, 57.3, -57.3]
reach left	[1.2, 0.6, -0.8]	[68.8, 34.4, -45.8]
folded	[0, -1.76, 1.74]	[0.0, -100.8, 99.7]

16 Summary

The project should be read as:

intact SO101 MuJoCo model + clean DH teaching model.

This keeps the robot visually recognizable and physically simulatable while still giving students a clear DH table, FK derivation, Jacobian, and IK solver.